YAT COLLEGE

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Solution Design

YAT LMS Cloud Architecture — Baseline Design

Engagement	YAT LMS Cloud Migration — Foundation Build
Document type	Technical design (Solution Design)
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Authored by	MTS Senior Architecture Team, in consultation with YAT ICT
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Implemented by	the AT2 Deployment Report (foundation build)
Classification	Internal — YAT ICT, and MTS personnel on signed MSA

Contents

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1. Purpose and Scope

This document specifies the baseline AWS architecture MTS will implement as the first build phase of the YAT LMS cloud migration. It translates the action plan approved at the close of the Business Case engagement into a concrete, implementable design. The design stops at "infrastructure ready for application deployment": the EC2 instance is provisioned with the OS, the RDS instance with an empty MySQL schema, and the ALB with placeholder health checks — no application binaries or production data are placed by the MTS build.

In scope of this design

- The production cloud foundation for the DOODLE LMS application.
- All compute, networking, identity, storage, database, autoscaling and monitoring needed to run the LMS as a multi-tier web workload in AWS.
- Single-region, single-Availability-Zone deployment in ap-southeast-2 (Sydney).

Out of scope — deferred to the follow-on HA design phase

- High-availability hardening (Multi-AZ database, cross-AZ compute resilience, failure-simulation testing).
- Disaster recovery to a second AWS region; DR runbook and tabletop testing.
- Application re-platforming (the LMS remains Windows Server 2016 + DOODLE + MySQL).

Out of MTS scope entirely — YAT ICT responsibility

- LMS application installation onto the EC2 instance(s) after handover.
- Database migration (extract from on-prem MySQL, load into RDS).
- Cutover — DNS switch, parallel running, decommissioning, user redirection.
- Organisational change management — CAB approvals, communications, training, post-cutover support.

2. Design Inputs and Requirements

2.1 Inputs

- LMS Application Specification — workload, SLAs, data footprint, integration points.
- LMS Cloud Migration Requirements — SLA, RPO/RTO targets.
- Engagement Role Brief — engagement scope, OS and application preservation.
- ICT Environment Overview and the On-Premises Network Diagram — the current state being migrated from.

2.2 Requirements the design must meet

Requirement	Target / note
Region / data residency	ap-southeast-2 (Sydney); all student PII within Australia (Privacy Act, APP 8)
Application stack	Preserved — Windows Server 2016 · MySQL · DOODLE
Concurrent users	200–300 typical; 500–700 peak during assessment windows
Data footprint	~178 GB, growing ~25 GB/year
Authentication	Preserve AD-LDAP integration over a private network to campus AD
High availability	Out of baseline scope — the 99.9% target is deferred to the follow-on HA

	design

3. Review of Existing Architecture

Not applicable — this is a greenfield cloud foundation; there is no existing cloud baseline to review. The on-premises current state is documented separately in the ICT Environment Overview.

4. Architecture Design

4.1 Assumptions and constraints

#	Assumption / constraint	Source
A1	Region must be ap-southeast-2 (Sydney) for data residency	LMS App Spec; Privacy Policy
A2	Application stack preserved: Windows Server 2016, MySQL, DOODLE	Engagement Role Brief
A3	Data footprint ~178 GB, +~25 GB/year (attachments + submissions dominate)	LMS App Spec
A4	Concurrent users 200–300 typical, 500–700 peak	LMS App Spec
A5	All student PII must remain within Australia	Privacy Act 1988, APP 8
A6	Preserve AD-LDAP integration over private network to campus AD	LMS App Spec
A7	Baseline is NOT high-availability; 99.9% deferred to the HA design	Scoping decision

4.2 AWS account and region

An AWS account scoped to the migration engagement, provisioned by YAT and shared with MTS for the build. Region ap-southeast-2 (Sydney); no deployment outside Australian regions. Availability Zone ap-southeast-2a for the baseline; the follow-on HA design introduces a second AZ.

4.3 Identity and Access Management (IAM)

Group	Purpose	Indicative permissions
YAT-ICT-Admins	YAT ICT day-to-day ops post-handover	Read-only on infra; full CloudWatch/RDS/EC2 console; no IAM changes
MTS-Consultants	MTS during build + support	Full admin during build; reduced post-handover
Application-Service	EC2 instance role for the LMS	RDS read/write; S3 read/write (attachments); CloudWatch logs
Read-Only-Auditors	Compliance / external auditors	Read-only on logs, metrics, configs

- MFA required for all YAT-ICT-Admins and MTS-Consultants users (Essential Eight; User Access Policy).
- No long-lived access keys for humans; programmatic access via IAM roles only; EC2 access via instance profile.

- Configuration decision left to the implementer: the MTS-Consultants permission boundary during build vs after handover.

4.4 Network topology

VPC 10.0.0.0/16 (room to expand), with DNS hostnames/resolution and VPC flow logs enabled. Single-AZ subnets in ap-southeast-2a:

Subnet	CIDR	Tier	Internet-facing?
public-web-a	10.0.1.0/24	Web / public load balancer	Yes
private-app-a	10.0.11.0/24	Application / LMS EC2	No
private-data-a	10.0.21.0/24	Database (RDS)	No

- Internet Gateway for public-subnet traffic; NAT Gateway in public-web-a for private-app outbound (Windows Update etc.).
- Route tables: public → IGW; private-app → NAT; private-data → no internet route.
- Connectivity to campus AD: Site-to-Site VPN (baseline choice); Direct Connect deferred unless latency requires it.
- The follow-on HA design adds the corresponding -b subnets in ap-southeast-2b.

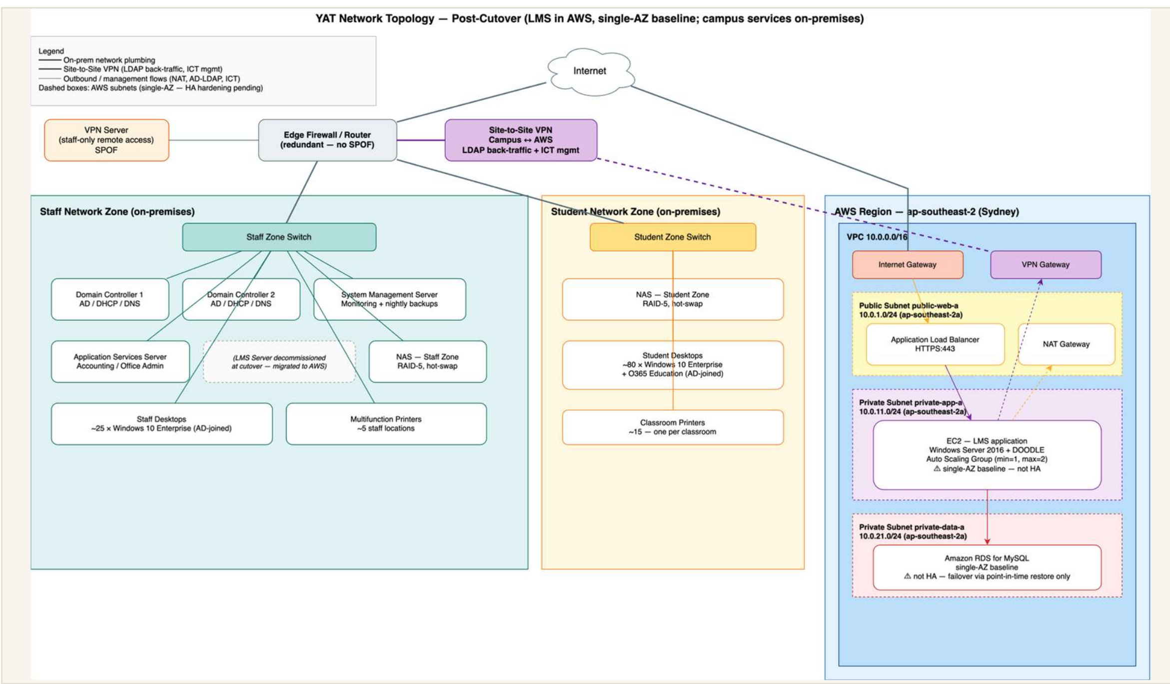

Figure 4.4 — LMS baseline network topology (single-AZ).

4.5 Compute (EC2 + Auto Scaling)

- EC2: general-purpose x86 (e.g. m6i.large — final type a C1 implementer decision); Windows Server 2016 AMI; placed in private-app-a (no public IP).
- EBS: gp3 root 100 GB + a gp3 data volume sized by the implementer (footprint + 12-month growth + 50% headroom).
- Auto Scaling Group: min 1 / desired 1 / max 2 (baseline); target-tracking on CPU at 70%; ELB+EC2 health checks; 300 s cooldown.
- The follow-on HA design expands the ASG (min 2, multi-AZ) and tunes scaling for assessment-window peaks.

4.6 Load balancing (ALB)

- Internet-facing ALB in public-web-a; HTTPS:443 listener forwarding to the LMS target group.
- Target group = the ASG instances; HTTP health check on the LMS health endpoint (30 s; 2 unhealthy → out of service).
- TLS via an ACM-issued certificate for the LMS DNS name (DNS strategy a C8 implementer decision).

4.7 Database (RDS)

- Amazon RDS for MySQL (preserves the existing data/schema); engine version confirmed against DOODLE at build time.
- General-purpose instance class (e.g. db.m6i.large — a C2 implementer decision); gp3 storage sized to footprint + growth.
- Multi-AZ DISABLED for the baseline (enabled in the HA design); storage encryption enabled (KMS).
- Placed in private-data-a; not publicly accessible; 7-day automated backups (window 22:00–04:00 AEST); maintenance Sun 02:00–06:00 AEST.
- Schema and data migration are YAT ICT's responsibility, not MTS's — MTS provisions an empty instance.

4.8 Storage

Resource	Type	Purpose
EC2 root volume	EBS gp3 100 GB	OS and LMS application install
EC2 data volume	EBS gp3 (sized by implementer)	LMS application data, attachments staging
Attachments bucket	S3	Course attachments + submissions; lifecycle to Glacier Deep Archive after 24 months
Backups bucket	S3	Off-instance mysqldump exports, file snapshots

- Both buckets: block all public access; SSE-S3 encryption; versioning enabled; access logging to a log bucket.

4.9 Security

Security group	Inbound	Outbound
sg-alb	HTTPS:443 from 0.0.0.0/0	HTTP/HTTPS to sg-app
sg-app	from sg-alb; RDP:3389 from MTS bastion (design left to implementer)	MySQL:3306 to sg-db; HTTPS via NAT; LDAP/LDAPS to campus AD
sg-db	MySQL:3306 from sg-app only	none

- Encryption in transit: HTTPS ALB→EC2; TLS EC2→RDS; AWS calls over TLS via VPC endpoints where available.
- Encryption at rest: EBS, RDS, and S3 (SSE-S3) all enabled.
- Operates under the AWS Shared Responsibility Model — AWS secures the cloud; YAT/MTS secure the OS, application, IAM, data and access in the cloud.

4.10 Monitoring (baseline)

Standard CloudWatch metrics for EC2, RDS, ALB and Auto Scaling. Baseline alarms (HA-tuned alarms come in the follow-on HA design):

Alarm	Threshold
EC2 CPU high	≥ 80% over 10 min
RDS CPU high	≥ 80% over 10 min
RDS free storage low	< 15%
ALB 5XX	> 10 / min
RDS connections high	> 80% of max_connections

- Logging: VPC flow logs and RDS logs → CloudWatch Logs (90-day retention); ALB access logs → S3; EC2 OS logs via the CloudWatch Agent.

4.11 Naming and tagging conventions

Naming pattern yat-lms-<resource-type>-<env>-<az-or-purpose> (e.g. yat-lms-alb-prod). Mandatory tags:

Tag	Value
Project	YAT-LMS-Migration
Environment	Production
Owner	YAT-ICT
ManagedBy	MTS-Migration during build → YAT-ICT post-handover
CostCentre	YAT-LMS
DataClassification	Confidential (PII) or Internal

4.12 Backup

Resource	Mechanism	Retention
RDS database	Automated daily backups + transaction logs	7 days
EC2 EBS volumes	Daily AMI snapshot (Data Lifecycle Manager)	14 days
LMS attachments (S3)	Versioning + lifecycle to Glacier Deep Archive	Versioned; archive after 24 months

- Cross-Region backup copies are out of scope for the baseline — addressed in the follow-on HA design.

4.13 Recovery objectives — baseline state

The baseline's recovery posture is backup-based: RPO is up to the last automated backup (daily) and RTO is restore-based (single-AZ restore). The HA targets (99.9% availability, RPO ≤ 1 h, RTO ≤ 4 h) are deliberately not met by this baseline and are the objective of the follow-on HA design.

4.14 Components requiring vertical scaling

RDS instance-class changes require a modify-and-apply (a brief interruption, taken in the maintenance window); EBS volumes support online resize with no downtime.

4.15 Single points of failure removed

Not applicable — this baseline is single-AZ by design; the single RDS instance and the single AZ are known single points of failure, deliberately deferred to the follow-on HA design.

4.16 Configuration decisions left to the implementer

The design is opinionated where it matters and silent where the implementer must show judgement. Each decision below is to be made and evidenced in the Deployment Report.

#	Decision	Why left open
C1	EC2 instance type (general-purpose family)	Size against the LMS workload
C2	RDS instance class (general-purpose family)	Size against the workload
C3	EBS data volume + RDS storage size	Compute from data footprint + growth
C4	ASG scaling threshold	Rationalise against the expected CPU profile
C5	MTS-Consultants permission boundary	Adapt to the lab access scope
C6	Bastion / jump-host design for RDP	Left to the implementer
C7	MySQL engine version	Confirm against the DOODLE compatibility matrix
C8	DNS strategy + ACM certificate domain	Confirm the LMS hostname with YAT ICT

5. Implementation Sequencing

Not applicable — this is a greenfield build in a new account, not a change to a running system; build order is at the implementer's discretion and there is no live service to sequence around or roll back.

6. Simulation and Verification Plan

Not applicable — failure/resize simulation and availability verification are the subject of the follow-on HA design and its deployment report; the baseline build is verified functionally per the Deployment Report's testing section.

7. Out of Scope

Stated explicitly so the implementer knows what not to build (these are the deliberate inputs to the follow-on HA design):

- Multi-AZ database; cross-AZ subnets (private-app-b, private-data-b); ASG capacity ≥ 2 across AZs.
- HA-tuned monitoring (cross-AZ latency, RDS replica lag); cross-Region backup copies and DR runbook.
- Failure-simulation testing; automated availability reporting against the 99.9% target.
- And, out of MTS scope entirely: LMS application install, data migration, cutover, and change management (YAT ICT).

8. References

- LMS Application Specification; LMS Cloud Migration Requirements; Engagement Role Brief.
- ICT Environment Overview; On-Premises Network Diagram.
- Privacy Policy; User Access Policy; Security and Incident Response; Industry Standards Reference (AWS Well-Architected, ACSC Essential Eight).

Document control

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Successor document	YAT LMS Cloud Architecture — HA Design (follow-on phase)